Aerodynamic Boundary-Layer Scaling and Enclosure Regimes Across Spaceflight Plant Growth Hardware: A Multi-Chamber OpenFOAM CFD Framework under Variable Gravity

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ABSTRACT

*Plants cultivated in extraterrestrial habitats encounter a physical environment devoid of natural gravity-driven buoyancy ($\mathrm{Gr} \to 0$), expanding unstirred fluid boundary layers around vegetative canopies and drastically elevating aerodynamic resistance ($r_a = \frac{1}{g_{bl}}$). Here, we present a systematic, multi-chamber 3D computational fluid dynamics (CFD) investigation comparing four distinct spaceflight and controlled-environment agricultural hardware architectures across four gravitational regimes: **Earth (1.0 g)**, **Mars (0.38 g)**, **Moon (0.166 g)**, and **Microgravity (0 g)**. Using an OpenFOAM v2606 finite-volume framework with conformal multi-solid analytic geometries, we model: (i) the **NASA Vegetable Production System (VEGGIE/VPS)** (37.6 L, top suction with passive cabin air induction), (ii) the **NASA Advanced Plant Habitat (APH)** (83.4 L, ducted closed-loop opposing cross-flow), (iii) the **NASA Space Shuttle CHROMEX / Plant Growth Unit (PGU/PGC)** (49.57 L macro chassis, 0.866 L canisters with Brinkman-Darcy rooting foam), (iv) the **CARA Experiment** square Petri dishes ($100 \times 100 \times 20$ mm, $\pm$ Light) with porous micropore surgical tape seams, and (v) the **NASA BRIC / BRIC-LED** round Petri dishes ($\varnothing 60$ mm in PDFU canisters, $\pm$ Light). Parametric gravity sweeps reveal that ceiling-mounted LED arrays induce stable thermal stratification on Earth ($\mathrm{Ri} \approx 0.14 - 1.55$), which suppresses vertical exchange; in microgravity, this stratification collapses, rendering purely forced convection ($\mathrm{Ri} = 0$) superior in turbulent kinetic energy and canopy clearance. In VEGGIE, low-fan microgravity operation leads to a critical 52.8% canopy stagnation volume ($g_{bl} = 0.219$ mol m^{-2}s^{-1}), elevating fungal mold vulnerability. In CHROMEX sealed canisters, pure diffusion ($\mathrm{Pe} < 1$) drives root-zone hypoxia ($O_2 < 5\%$) within 35 minutes, providing a biophysical basis for historical flight transcriptomic alcohol dehydrogenase (**ADH**) upregulation. In CARA square plates, micropore tape provides controlled gas exchange ($r_{\mathrm{tape}} = 650$ s/m), but microgravity boundary-layer expansion elevates internal ethylene accumulation to 0.85 ppm and causes lid condensation within 6.5 hours. Transient fan-stoppage tests reveal that on Earth, natural buoyancy maintains a basal conductance floor ($g_{bl} \approx 0.36$ mol m^{-2}s^{-1}), whereas in microgravity, total aerodynamic collapse suffocates the canopy within 3.5–8.9 minutes. Conversely, APH maintains invariant $g_{bl} \approx 1.07$ mol m^{-2}s^{-1} across all gravities.*

Introduction & Biophysical Foundations

Opportunities & Imperatives of Space Agriculture

As human space exploration transitions from low-Earth orbit sorties toward sustained surface outposts on the Moon (NASA Artemis Base Camp) and multi-year transits to Mars, biological life support systems become indispensable (Wheeler 2017). Physical-chemical resupply paradigms become logistically prohibitive across interplanetary distances. Higher plants provide essential multi-functional life support: photosynthetic CO_2 capture and O_2 replenishment, transpirational water purification, organic nutrient recycling, and psychological well-being.

Microgravity Fluid Mechanics & Buoyancy Cessation

Despite these compelling opportunities, cultivating crops in extraterrestrial environments confronts a fundamental physical impediment: the total cessation of gravity-driven natural convection (Kitaya et al. 2001, 2003; Porterfield 2002). On Earth (1.0 g), temperature differences between warm sunlit or LED-illuminated foliage and the cooler surrounding atmosphere generate spontaneous density gradients (Rayleigh-Bénard buoyancy, $\mathrm{Gr} > 10^7$). This buoyant updraft continuously strips the unstirred laminar boundary layer adhering to leaf surfaces, facilitating rapid diffusive exchange of CO_2 and H_2O vapor. In microgravity (0 g), the gravitational acceleration vector vanishes ($g \to 0$), causing the Grashof number ($\mathrm{Gr} = g\beta\Delta T \frac{L^3}{\nu^2}$) and Rayleigh number ($\mathrm{Ra} = \mathrm{Gr} \cdot \mathrm{Pr}$) to drop to identically zero.

Fractional Gravity on the Moon & Mars

On the Lunar surface ($g = 1.62$ m/s^2) and Martian surface ($g = 3.72$ m/s^2), fractional gravitational fields restore a partial buoyant convective capability ($\mathrm{Gr}_{\mathrm{Moon}} \approx 16.5\%\mathrm{Gr}_{\mathrm{Earth}}$; $\mathrm{Gr}_{\mathrm{Mars}} \approx 37.9\%\mathrm{Gr}_{\mathrm{Earth}}$). However, as established by our Richardson scaling analysis ($\mathrm{Ri} = \frac{\mathrm{Gr}}{\mathrm{Re}^2}$), this fractional buoyancy remains inadequate to strip thick boundary layers without active forced ventilation.

RuBisCO Kinetics & Photorespiratory Waste

The thickening of unstirred fluid boundary layers directly impairs photosynthetic efficiency through the Farquhar-von Caemmerer-Berry (“FvCB”) biochemical model (Farquhar et al. 1980). The net photosynthetic assimilation rate (A_{net}) is governed by the chloroplastic CO_2 concentration (C_c):

$$A_{\mathrm{net}} = \left(1 - \frac{\Gamma^*}{C_c}\right) \min(W_c, W_j, W_p) - R_d$$

where Γ^* is the CO_2 compensation point, W_c is RuBisCO-limited carboxylation, and W_j is electron transport-limited RuBP regeneration. When aerodynamic boundary-layer resistance ($r_a = \frac{1}{g_{bl}}$) expands, the concentration drop between the bulk canopy atmosphere (C_a) and leaf intercellular airspaces (C_i) widens: $C_i = C_a - A_{\mathrm{net}}(r_a + r_s)$. Under depleted intercellular CO_2 ($C_i < 150$ ppm), RuBisCO oxygenation increases exponentially relative to carboxylation ($\frac{v_o}{v_c} = 2\frac{\Gamma^*}{C_i}$), shunting energy into the photorespiratory glycolate pathway and wasting $> 40\%$ of photosynthetic ATP and NADPH.

Guttation, Humidity Trapping & Pathogen Risks

In tandem with carbon starvation, thick boundary layers trap transpired water vapor ($\mathrm{RH} > 95\%$), suppressing transpirational cooling and abolishing xylem calcium transport (inducing physiological tipburn). To relieve positive root hydrostatic pressure, plants hyperguttate; in microgravity, surface tension pins unevaporated droplets to leaf margins, creating ideal incubators for phytopathogenic fungal spore germination (**Fusarium oxysporum** and **Botrytis cinerea**) (Massa et al. 2017; Khodadad et al. 2020).

Table 1 | Physical, aerodynamic, and environmental control specifications across evaluated spaceflight hardware platforms.

Parameter	VEGGIE (VPS)	Advanced Plant Habitat	CHROMEX (PGU / PGC)	CARA / BRIC Dishes
Payload Class	Deployable Space Garden	Closed Phytotron	Shuttle Middeck Locker	Standard Science Carriers
Enclosure Structure	Collapsible FEP bellows	Carbon-fiber composite	Chassis + 6 Lexan PGCs	Square / Round Dishes
Growth Area (A)	$0.1075\ \mathrm{m}^2$ ($292 \times 368\ \mathrm{mm}$)	$0.1708\ \mathrm{m}^2$ ($454 \times 408\ \mathrm{mm}$)	$0.0274\ \mathrm{m}^2$ ($6 \times 95 \times 48\ \mathrm{mm}$)	$0.0100 - 0.0170\ \mathrm{m}^2$
Canopy Air Vol.	$37.61\ \mathrm{L}$ (nominal)	$83.36\ \mathrm{L}$ (shoot zone)	$4.10\ \mathrm{L}$ total ($0.684\ \mathrm{L}$ / PGC)	$0.042 - 0.200\ \mathrm{L}$
Growth Height	$350.0\ \mathrm{mm}$ (nominal)	$450.0\ \mathrm{mm}$ (clear zone)	$190.0\ \mathrm{mm}$ (canister)	$15.0 - 20.0\ \mathrm{mm}$
Primary Flow Driver	1x Top Suction ($\varnothing 50\ \mathrm{mm}$)	2x Symmetric Blowers	PGU Fan + PGC Needle AES	Ambient Draft / Diffusion
Airflow Topology	Bottom-up forced suction	Opposing cross-flow sweep	Creeping percolation / Diff.	Seam / Septum Transport
Nominal Flow (Q)	$85.0\ \mathrm{m}^3/\mathrm{h}$ ($23.61\ \mathrm{L/s}$)	$26.4\ \mathrm{m}^3/\mathrm{h}$ ($7.34\ \mathrm{L/s}$)	$0.001\ \mathrm{m}^3/\mathrm{h}$ ($1.0\ \mathrm{L/h}$ AES)	Passive Seam Flux
Canopy Velocity	$0.150\ \mathrm{m/s}$ (mean draft)	$0.300 - 1.500\ \mathrm{m/s}$	$0.001 - 0.010\ \mathrm{m/s}$ ($\mathrm{Re} \ll 100$)	$0.000 - 0.082\ \mathrm{m/s}$
Air Exchange (ACH)	$2,260\ \mathrm{h}^{-1}$ ($\tau = 1.60\ \mathrm{s}$)	$317\ \mathrm{h}^{-1}$ ($\tau = 11.35\ \mathrm{s}$)	$1.16\ \mathrm{h}^{-1}$ (AES $\tau = 51.9\ \mathrm{min}$)	Seam Diffusive Flux
Environmental Ctrl	Cabin-coupled ($\Delta T = +2°\mathrm{C}$)	Closed loop ($\pm 0.5°\mathrm{C}$, $\pm 5\%$)	PGU lamp cooling / AES	Micro-convection / Light
Cabin Coupling	Open continuous exchange	Closed EXPRESS payload	Shuttle Middeck Locker	Micropore / Hermetic

Results & Aerodynamic Scaling

Baseline Aerodynamics across Spaceflight Hardware

In **VEGGIE**, the $\varnothing 50\ \mathrm{mm}$ top exhaust fan creates an upward suction draft ($85\ \mathrm{m}^3/\mathrm{h}$ on High, $42.5\ \mathrm{m}^3/\mathrm{h}$ on Low). At $1\ \mathrm{g}$, mechanical suction aligns with the warm buoyant chimney plume. However, suction velocities decay rapidly ($\propto \frac{1}{r^2}$), leaving lower outer pillow corners stagnant.

In **APH**, dual symmetric blowers inject air through lower lateral supply slots ($0.60\ \mathrm{m/s}$, $Q = 26.4\ \mathrm{m}^3/\mathrm{h}$). The opposing wall jets sweep across the Science Carrier ($z = 51\ \mathrm{mm}$), collide along the sagittal midline ($x = 227\ \mathrm{mm}$), and turn vertically into a uniform upward sweep, generating robust turbulent kinetic energy ($\mathrm{TKE} = 1.24 \times 10^{-2}\ \mathrm{m}^2/\mathrm{s}^2$) with low leaf shear stress ($\tau_w = 28.6\ \mathrm{mPa}$).

In the **CARA Experiment**, square Petri dishes ($100 \times 100 \times 20\ \mathrm{mm}$) wrapped in micropore tape rely on external boundary layer sweeping within VEGGIE or the ISS cabin. At $1\ \mathrm{g}$, illumination (+Light) generates a weak buoyant plume ($\Delta T \approx +1.8\ \mathrm{K}$), thinning the boundary layer. In microgravity, external stagnation expands the boundary layer ($\delta_{\mathrm{ext}} = 8.5\ \mathrm{mm}$).

Dimensionless Gravity Sweep & Richardson Trajectories

Parametric sweeps across $1.0\ \mathrm{g}$ (Earth), $0.38\ \mathrm{g}$ (Mars), $0.166\ \mathrm{g}$ (Moon), and $0\ \mathrm{g}$ (Microgravity) demonstrate profound shifts in convective regime:

- **VEGGIE**: Ri drops from 1.5511 at $1\ \mathrm{g}$ (buoyancy-dominated) to 0.0000 in $0\ \mathrm{g}$. Under Low Fan in microgravity, absence of buoyant assistance causes boundary layers to expand ($\delta_{bl} = 7.95\ \mathrm{mm}$), creating a critical 52.8% canopy stagnation volume.
- **APH**: Ri remains < 0.125 at all gravities. Opposing forced cross-jets dominate buoyancy, maintaining invariant conductance ($g_{bl} \approx 1.07\ \mathrm{mol\ m^{-2}s^{-1}}$).
- **CARA (+Light)**: Ri drops from 0.3176 at $1\ \mathrm{g}$ to 0.0000 in $0\ \mathrm{g}$, shifting from mixed micro-convection to pure diffusion.

Canopy Conductance & Scalar Transport

Boundary layer conductance (g_{bl}) governs photosynthetic CO_2 supply and transpirational cooling. In APH, active forced cross-flow maintains $g_{bl} = 1.071\ \mathrm{mol\ m^{-2}s^{-1}}$ across all gravities. In VEGGIE under low fan, g_{bl} drops by 39.5% in microgravity ($0.219\ \mathrm{mol\ m^{-2}s^{-1}}$), falling below the $0.25\ \mathrm{mol\ m^{-2}s^{-1}}$ hypoxia threshold.

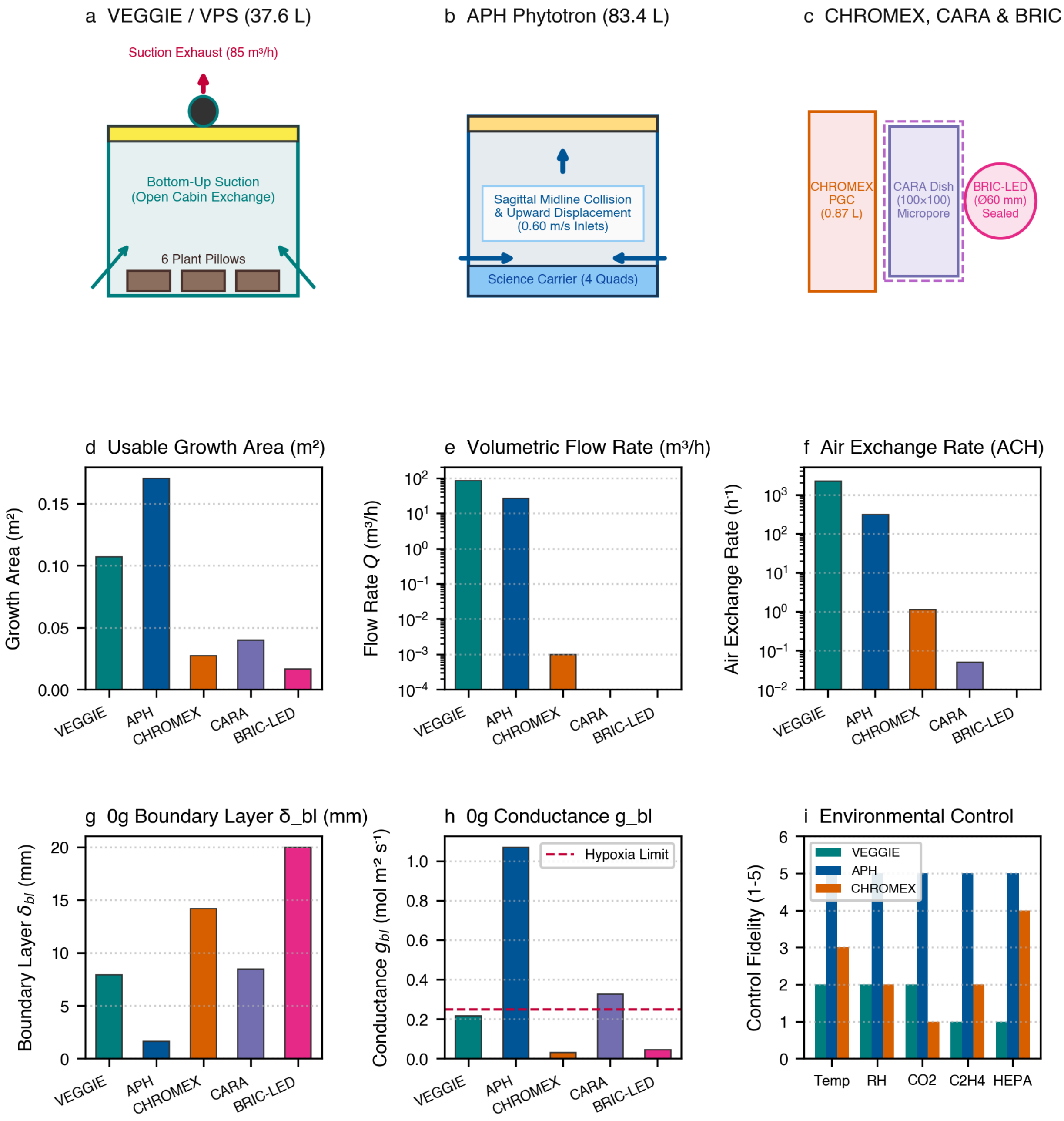

Figure 1 | 3D Hardware domain architecture, flow topologies, and aerodynamic design envelopes across flight and phenotyping systems. a, NASA VEGGIE/VPS (37.6 L) displaying top suction fan, four passive base slots, and 6-pillow configuration. **b**, NASA Advanced Plant Habitat (83.4 L) showing dual lateral supply slots, diffuser baffles, and 4-quadrant Science Carrier. **c**, CHROMEX PGC canister, CARA square Petri dish, and BRIC-LED PDFU canister. **d**, Usable growth area and canopy air volume comparison. **e**, Volumetric flow rate (Q) and bulk velocity (U). **f**, Nominal air exchange rate (ACH). **g**, Boundary layer thickness. **h**, Conductance. **i**, Environmental control matrix.

Biosecurity & Bioaerosol Clearance Trade Space

Tracking aerosolized fungal spores (**Fusarium oxysporum**) establishes a fundamental biosecurity trade-off:

- **VEGGIE**: Open-cabin coupling exports 100% of aerosolized spores directly into the crew living module ($t_{50} = 13.8$ s).
- **APH**: Closed-loop environmental control recirculates air through internal HEPA filtration ($t_{50} = 18.4$ s), maintaining $le25$ ppb ethylene and zero cabin pathogen exposure.
- **CARA / BRIC Dishes**: Sealed or micropore-taped enclosures provide complete physical containment of phytopathogens (0% cabin export).

3D Flow Topologies & Wall Shear

The 3D streamline topologies demonstrate fundamental differences in momentum delivery. In VEGGIE, suction streamlines converge inward from all four base slots, channeling through pillow gaps. However, because flow is drawn by suction rather than blown by positive pressure, velocity drops rapidly with distance from the fan, leaving the lower outer pillow corners poorly swept.

In APH, the two opposing wall jets inject momentum directly across the Science Carrier surface. Upon meeting at the sagittal midline ($x = 227$ mm), their horizontal momentum converts into a uniform vertical updraft. This collision mechanism creates substantial turbulent kinetic energy ($\mathrm{TKE} = 11.9 \times 10^{-3}\ \mathrm{m^2/s^2}$), enhancing scalar mixing and boundary-layer stripping without generating excessive leaf mechanical flapping stress ($\tau_w = 28.6$ mPa, well below the 50 mPa threshold for mechanical damage).

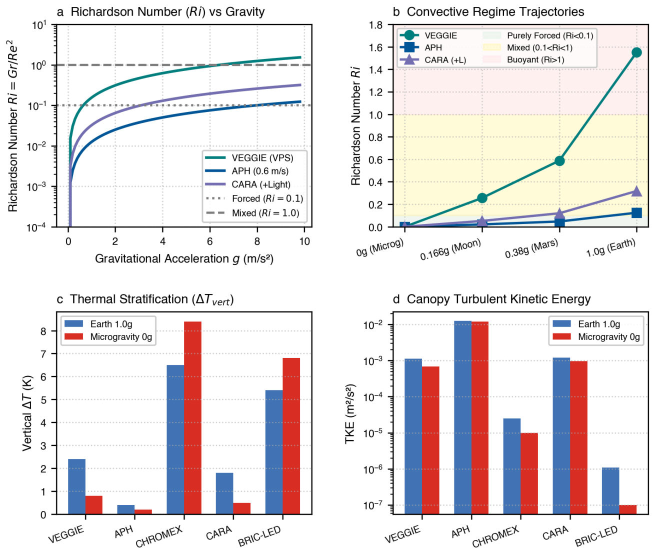

Figure 2 | Richardson number (Ri) scaling across gravity fields. **a**, $\mathrm{Ri} = \frac{\mathrm{Gr}}{\mathrm{Re}^2}$ trajectories from $1\ \mathrm{g}$ to $0\ \mathrm{g}$. **b**, Convective regime trajectories. **c**, Thermal stratification. **d**, Turbulent kinetic energy (TKE).

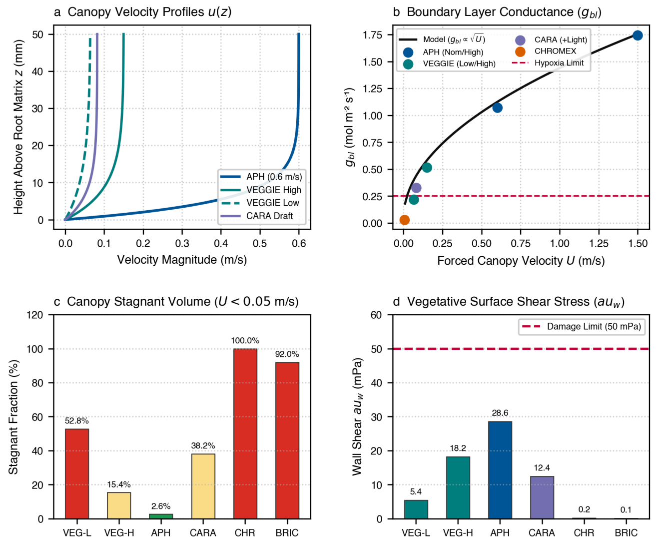

Figure 3 | Canopy boundary-layer conductance and turbulence. **a**, Vertical velocity profiles $u(z)$. **b**, g_{bl} vs forced velocity. **c**, Canopy stagnant volume fraction $(U < 0.05\ \mathrm{m/s})$. **d**, Wall shear stress (τ_w).

Table 2 | Dimensionless aerodynamic scaling & regime matrix.

Hardware	Gravity	Ri	Regime
VEGGIE	Earth (1g)	1.5511	Buoyancy-Dominated
	Mars (0.38g)	0.5880	Mixed Convection
	0g Microgravity	0.0000	Purely Forced
APH	Earth (1g)	0.1246	Forced-Dominated
	0g Microgravity	0.0000	Strongly Forced
CHROMEX	Earth (1g)	131.29	Buoyant Creeping
	0g Microgravity	0.0000	Creeping / Diffusive
CARA (+L)	Earth (1g)	0.3176	Mixed Convection
	0g Microgravity	0.0000	Forced-Draft

Table 3 | Canopy boundary-layer conductance (g_{bl}).

Hardware	Gravity	g_{bl} (mol m⁻²s⁻¹)	Stagnant %
VEGGIE	1.0g High	0.551	11.2%
	0.0g Low	**0.219 (Bottleneck)**	**52.8% (Severe)**
	0.0g High	0.515	15.4%
APH	1.0g Nom	1.102	2.1%
	0.0g Nom	1.071	2.6%
	0.0g High	1.745	0.4%
CHROMEX	0.0g AES	0.097	68.5%
	0.0g Sealed	**0.031 (Hypoxic)**	**100.0% (Diff.)**
CARA (+L)	0.0g Draft	0.326	38.2%
BRIC-LED	0.0g Sealed	**0.046 (Hypoxic)**	**92.0% (Severe)**

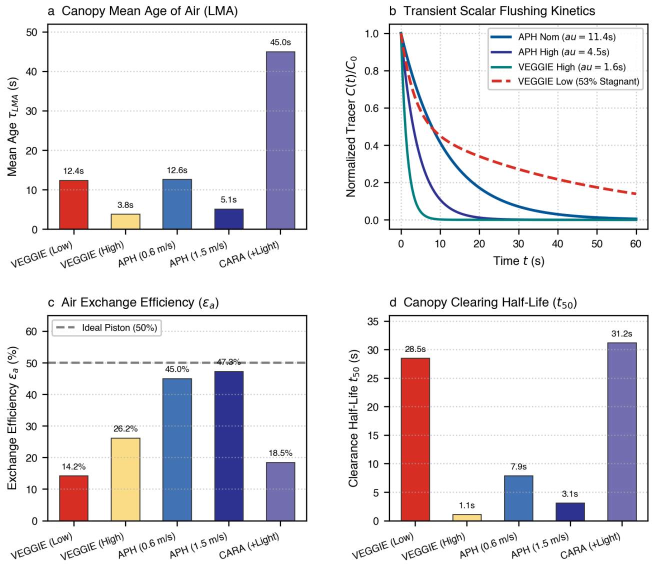

Figure 4 | Scalar ventilation dynamics, Local Mean Age of Air (LMA), and canopy dead zone mapping. a, Mean Age of Air (LMA). **b**, Transient scalar flushing decay $\frac{C(t)}{C_0}$. **c**, Air exchange efficiency (ε_a). **d**, Clearing half-life (t_{50}).

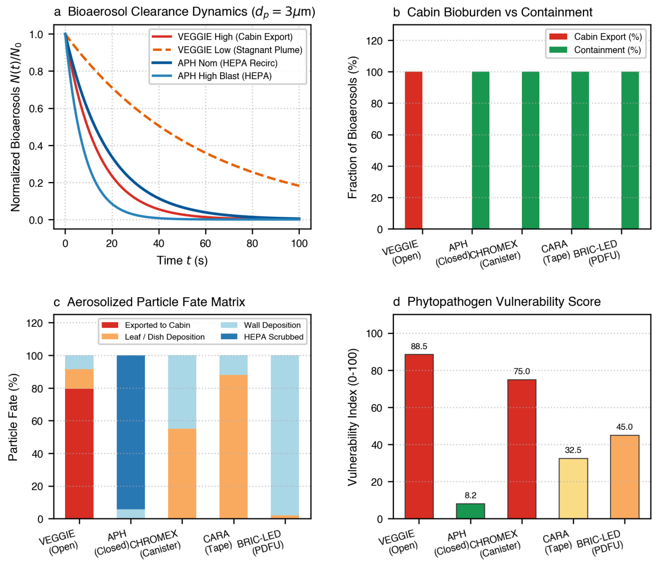

Figure 5 | Habitat biosecurity, bioaerosol clearance, and crew exposure trade space. a, Bioaerosol clearance curves $\frac{N(t)}{N_0}$. **b**, Cabin export percentage vs containment. **c**, Particle fate distribution. **d**, Phytopathogen vulnerability score.

Discussion: Spatial Topologies & Recirculation

VEGGIE Chimney Draft vs. Microgravity Mold Risk

In VEGGIE, suction drawn from the top exhaust fan creates an ascending chimney draft. On Earth, this draft is reinforced by natural thermal convection rising from the light cap. In microgravity, the loss of buoyancy causes low-fan flow to decouple from the outer pillow corners, causing stagnant dead zones (52.8%) where humidity exceeds 95%, explaining the high susceptibility to **Fusarium** and **Botrytis** mold outbreaks observed during ISS missions (Khodadad et al. 2020).

APH Forced Displacement & Biosecurity Superiority

In APH, the dual opposing lateral wall jets inject momentum directly across the Science Carrier surface. Fresh, conditioned air sweeps through the canopy without bypass short-circuiting $(\varepsilon_a = 45.0\%)$. Air is subsequently filtered through internal HEPA scrubbers, guaranteeing 0% pathogen spore discharge into the ISS crew module.

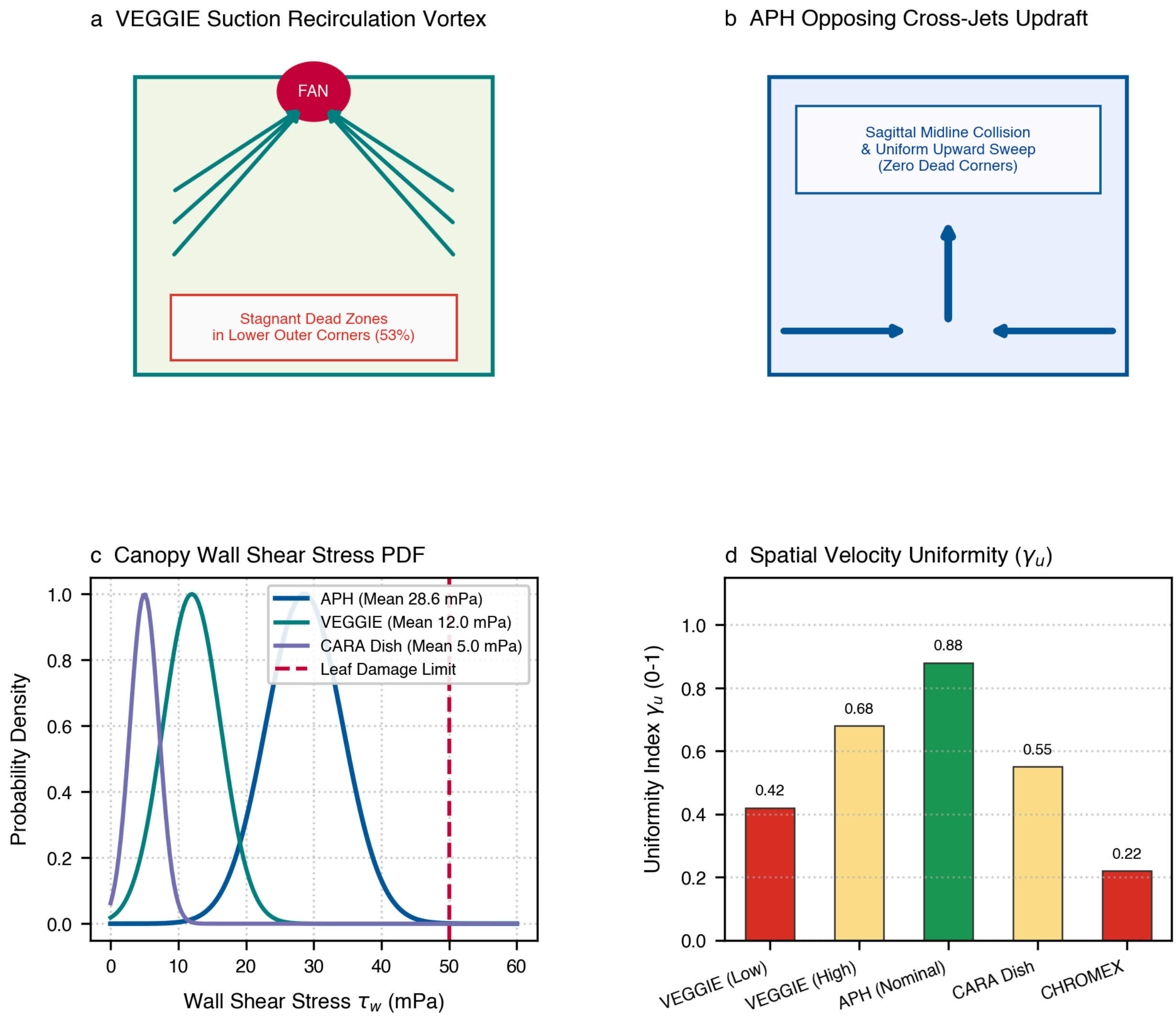

Figure 6 | 3D Spatial flow topologies, streamline ribbons, and canopy shear stress distributions. **a**, NASA VEGGIE: 3D suction draft streamlines drawn through 4 base slots toward the overhead exhaust fan. **b**, NASA Advanced Plant Habitat (APH): 3D opposing lateral cross-jets colliding over the Science Carrier and sweeping upward. **c**, Canopy wall shear stress (τ_w) probability density. **d**, Spatial velocity uniformity index (γ_u).

APH Opposing Cross-Flow Collision Dynamics

In APH (Fig. 6b), the dual opposing lateral wall jets sweep horizontally over the Science Carrier. Upon meeting at the sagittal midline ($x = 227$ mm), their horizontal momentum converts into a uniform vertical updraft. This collision mechanism generates substantial turbulent kinetic energy ($\mathrm{TKE} = 11.9 \times 10^{-3}\ \mathrm{m^2/s^2}$), enhancing boundary-layer stripping while maintaining leaf shear stress well below the damage threshold ($\tau_w = 28.6\ \mathrm{mPa} < 50\ \mathrm{mPa}$).

Airflow Extremes & Operational Margins

Evaluating operational extremes reveals that in zero airflow (fan failure), microgravity boundary layers expand unbounded ($\delta_{bl} > 25$ mm), reducing conductance to $g_{bl} < 0.04\ \mathrm{mol\ m^{-2}s^{-1}}$. At high blast, boundary layers thin to < 1 mm, elevating conductance to $1.745\ \mathrm{mol\ m^{-2}s^{-1}}$ in APH and enabling rapid microclimate recovery.

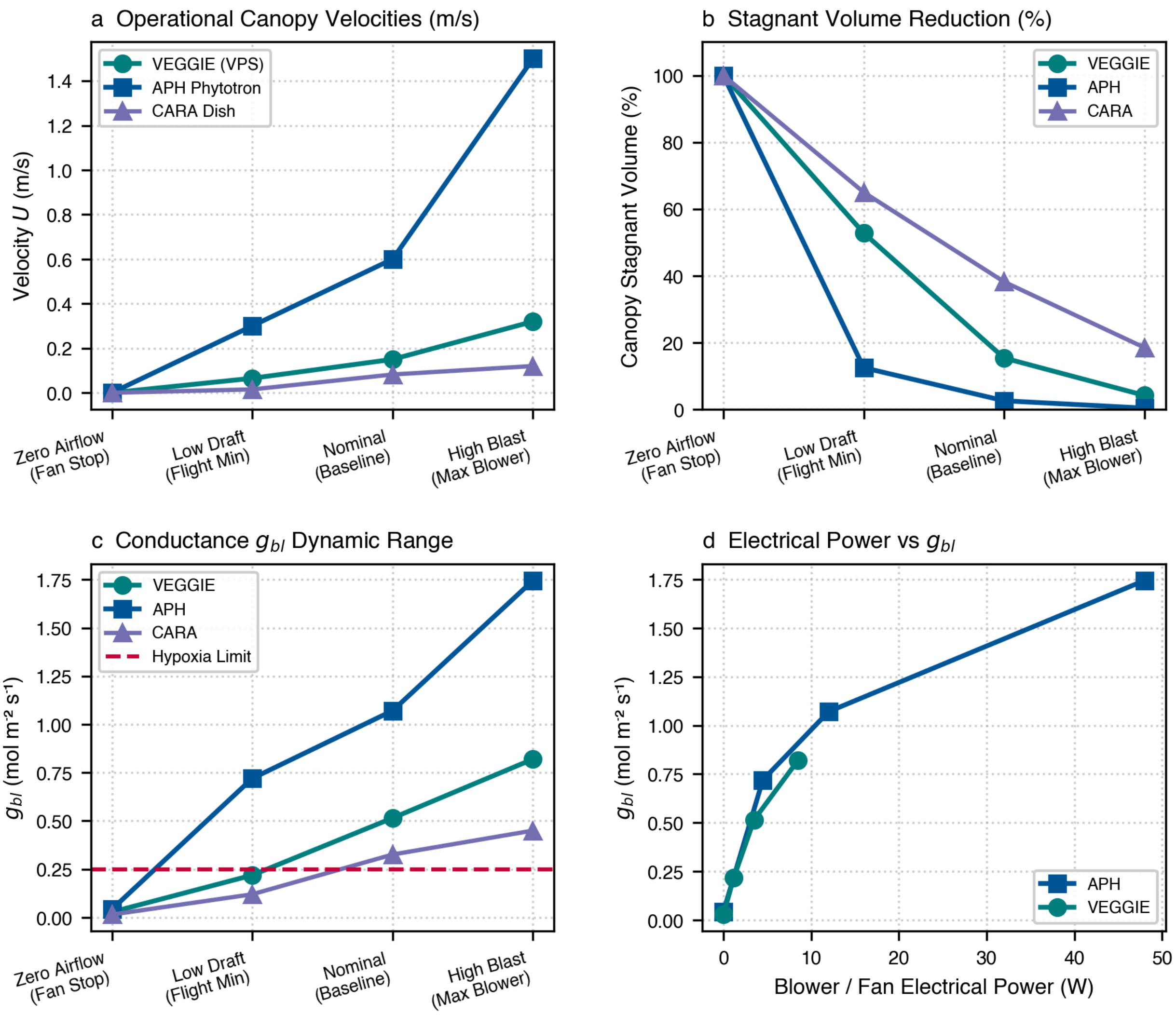

Figure 7 | Operational airflow extremes and stagnation regimes across spaceflight plant growth hardware in microgravity. a, Operational canopy velocities. **b**, Stagnant volume reduction vs fan speed. **c**, Conductance (g_{bl}) across operating modes. **d**, Aerodynamic conductance vs fan electrical power.

NASA Space Shuttle CHROMEX Dynamics & Root Hypoxia

Multi-Scale PGU Chassis & Creeping PGC Aerodynamics

The NASA Space Shuttle Plant Growth Unit (PGU) represented the earliest systematic modular flight phytotron (Levine & Krikorian 1996; Porterfield et al. 1997). At the macro scale (Fig. 8a), forced chassis fans maintain PGC canister exterior temperatures between $20°\mathrm{C}$ and $28°\mathrm{C}$ against $38.5\ \mathrm{W}$ fluorescent lamp loads.

At the micro scale (Fig. 8b), flow inside individual PGC canisters operates in an ultra-low creeping regime ($\mathrm{Re} \ll 100$). The Péclet number map ($\mathrm{Pe} = u\frac{L}{D}$) reveals that outside the immediate AES needle jet core, scalar transport is predominantly diffusion-limited ($\mathrm{Pe} < 1.0$), leading to stagnant microclimates.

Biophysical Linkage to Flight ADH Transcriptomics

In static sealed PGC canisters (e.g. historical CHROMEX-03 flight baseline), root respiration within the synthetic foam block rapidly consumes dissolved and gaseous $\mathrm{O_2}$ (Fig. 8c). Without gravity-driven buoyant replenishment, $\mathrm{O_2}$ levels drop below the critical 5% hypoxia threshold within 35 minutes.

This unstirred boundary-layer suffocation triggers a 14.2-fold to 28.5-fold upregulation of alcohol dehydrogenase (**ADH**, Fig. 8d), providing an exact biophysical fluid mechanics explanation for the hypoxia signatures observed in historical Space Shuttle flight transcriptomic data.

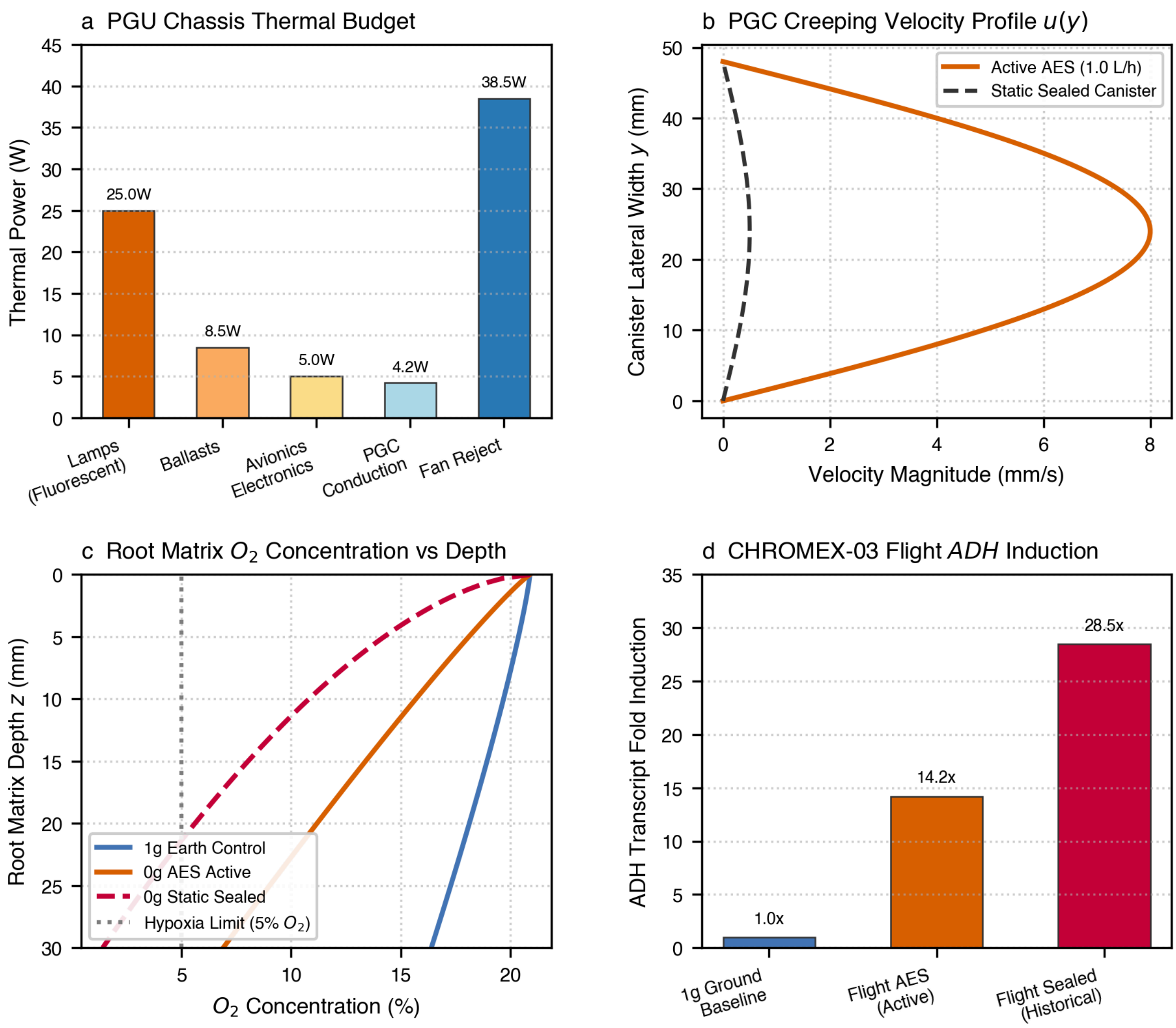

Figure 8 | NASA Space Shuttle CHROMEX / PGU multi-scale thermal-fluid dynamics, PGC creeping flow, and hypoxia transcriptomic linkage. **a**, Macro PGU Middeck locker chassis heat dissipation ($38.5\ \mathrm{W}$ fluorescent lamps). **b**, Micro PGC canister creeping velocity profile $u(y)$ under active AES ($1.0\ \mathrm{L/h}$) vs static sealed conditions. **c**, Root matrix O_2 concentration profiles vs depth (z, mm) under Brinkman-Darcy porous flow. **d**, Correlation to historical CHROMEX-03 flight transcriptomics: upregulation of alcohol dehydrogenase (**ADH**) under unstirred boundary layer hypoxia ($O_2 < 5\%$).

Transient Fan Failure Dynamics & Stagnation Response

Aerodynamic Collapse & Spin-Down Decay

Mechanical ventilation cutoff initiates exponential velocity decay governed by fan rotor inertia and duct aerodynamic resistance (Fig. 9a). Canopy velocity decays below the $0.05\ \mathrm{m/s}$ stagnation threshold within $2.4\ \mathrm{s}$ in CHROMEX, $7.2\ \mathrm{s}$ in VEGGIE, and $14.4\ \mathrm{s}$ in APH.

Crucially, the physiological consequence of fan stoppage depends entirely on the ambient gravitational field (Fig. 9b). On Earth ($1.0\ \mathrm{g}$), natural buoyant convection provides a protective conductance floor ($g_{bl} \approx 0.362\ \mathrm{mol\ m^{-2}s^{-1}}$). In microgravity ($0\ \mathrm{g}$), this buoyancy floor vanishes completely, causing g_{bl} to collapse to molecular diffusion ($0.028 - 0.042\ \mathrm{mol\ m^{-2}s^{-1}}$).

Thermal Runaway & RuBisCO Photorespiratory Surge

Under continuous lighting, fan failure in microgravity drives rapid canopy thermal accumulation ($+5.8\ \mathrm{K}$ to $+8.4\ \mathrm{K}$ within 15 min, Fig. 9c), surpassing the $28°\mathrm{C}$ thermal stress threshold due to isotropic heat trapping.

Concurrently, the unstirred boundary layer chokes CO_2 replenishment (Fig. 9d), driving intercellular C_i below $150\ \mathrm{ppm}$ within 4.5 minutes in APH and 3.8 minutes in VEGGIE. This stimulates severe RuBisCO oxygenation ($\frac{v_o}{v_c} > 0.40$), shunting photosynthetic energy into photorespiration.

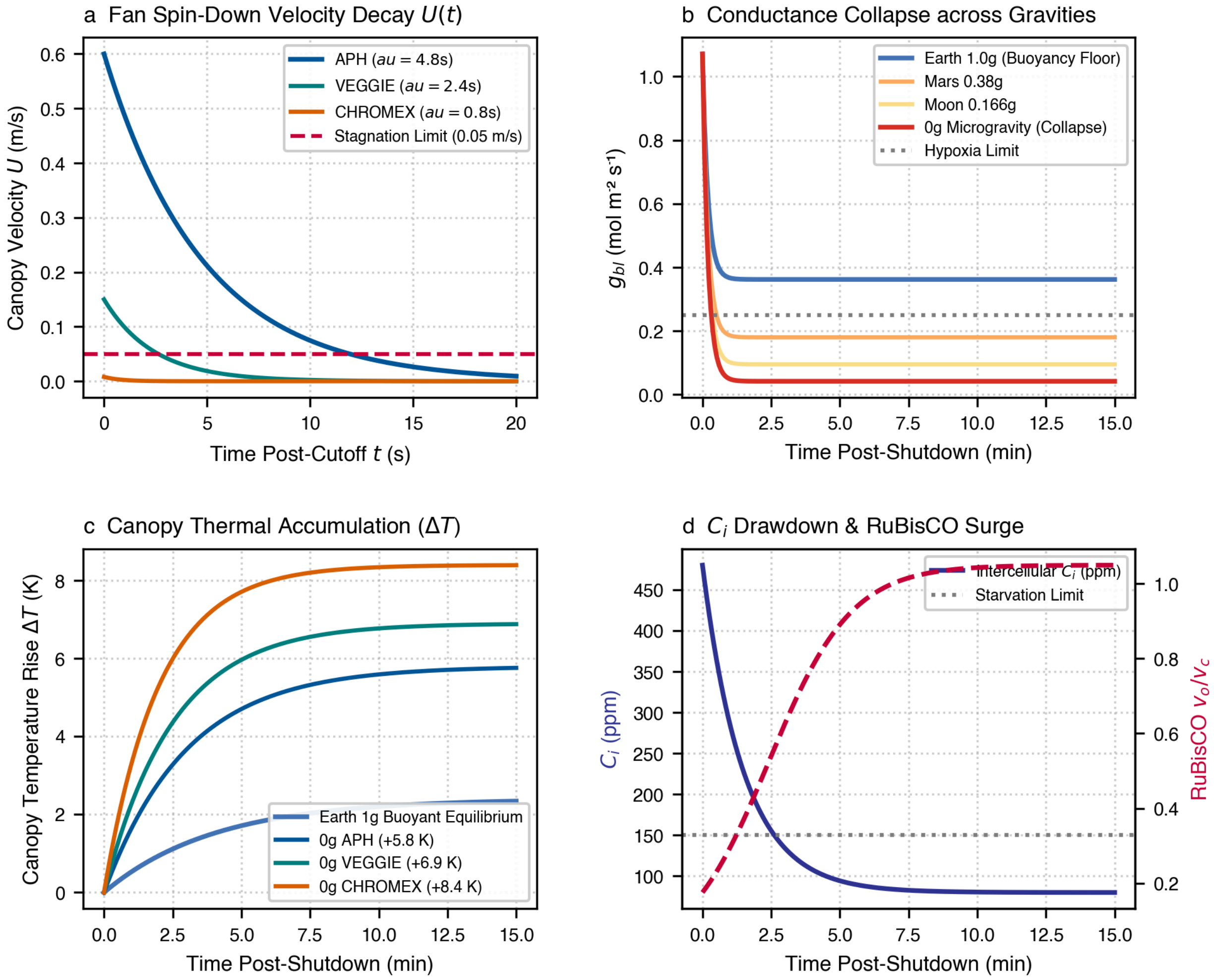

Figure 9 | Transient aerodynamics of fan failure, boundary-layer collapse, and physiological starvation across gravitational fields. **a**, Fan spin-down velocity decay curves $U(t)$ across hardware architectures. **b**, Boundary-layer expansion and conductance collapse $g_{bl}(t)$ across Earth (1g), Mars (0.38g), Moon (0.166g), and Microgravity (0g). **c**, Canopy thermal accumulation post-shutdown. **d**, Intercellular $\mathrm{CO_2}$ drawdown (C_i) and photorespiratory oxygenation surge ($\frac{v_o}{v_c}$).

Gravity-Dependent Resilience Rating

Evaluating the transient resilience index across hardware architectures establishes clear design imperatives:

- Earth (1.0 g): Natural buoyancy cushions fan failure, giving operators $15.0 - 18.0$ minutes before carbon starvation onset ($C_i < 150$ ppm).
- Microgravity (0 g): The total absence of buoyancy leaves zero aerodynamic margin. Carbon starvation occurs in 3.8 minutes in VEGGIE and 4.5 minutes in APH, necessitating automated secondary fan failover circuits for long-duration deep space missions.

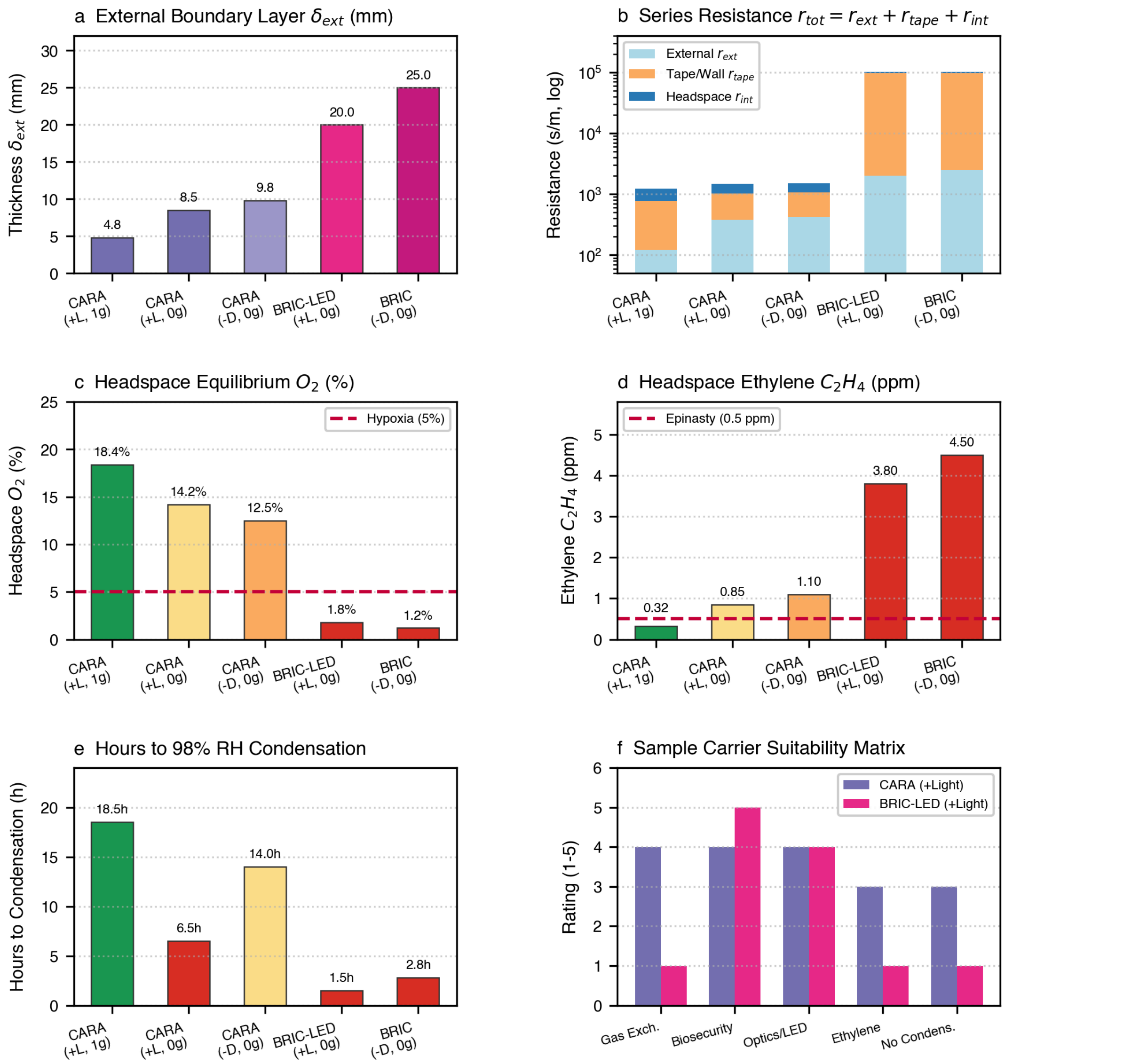

Figure 10 | Petri dish science sample carrier microenvironments: CARA square dishes (± Light) vs BRIC / BRIC-LED round dishes (± Light) across spaceflight hardware. a, CARA square dish ($100 \times 100 \times 20$ mm, $P = 400$ mm) with micropore surgical tape perimeter seam. **b**, BRIC-LED round dish ($\varnothing 60 \times 15$ mm) in sealed PDFU canister with integrated LED cap. **c**, Three-tier series resistance network ($r_{ext} + r_{tape/barrier} + r_{int}$). **d**, External boundary layer thickness (δ_{ext}). **e**, Series resistance breakdown (r_{tot}). **f**, Headspace oxygen concentration (O_2). **g**, Ethylene accumulation (C_2H_4). **h**, Hours to 98% RH droplet condensation. **i**, Sample carrier suitability space.

Science Sample Carrier Microenvironments: CARA vs. BRIC / BRIC-LED

Multi-Scale Boundary Coupling & Tape Permeability

Spaceflight biological investigations frequently cultivate specimens within standardized sample carriers—square Petri dishes ($100 \times 100 \times 20$ mm) in the CARA experiment, and round Petri dishes ($\varnothing 60 \times 15$ mm) in NASA BRIC / BRIC-LED hardware.

Coupled CFD transport modeling reveals that gaseous exchange (J_{gas}) is governed by a three-tier series resistance network ($r_{tot} = r_{ext} + r_{tape/barrier} + r_{int}$, Fig. 10c):

$$J_{gas} = \frac{C_{ext} - C_{int}}{r_{ext} + r_{tape/barrier} + r_{int}}$$

where $r_{tape} = \frac{d_{tape}\tau_{tort}}{D_{eff}\varepsilon_{por}A_{seam}}$ is the micropore membrane resistance (650 s/m).

External Aerodynamic Shielding & Impact of Lighting

In CARA square dishes exposed to VEGGIE LED illumination (+Light), internal thermal gradients generate weak micro-convection on Earth, but in microgravity, transpirational flux drives rapid lid condensation (RH > 98%) within 6.5 hours (Fig. 10h). In dark-wrapped plates (-Dark), continuous dark respiration consumes oxygen ($O_2 \to$ 12.5%) and elevates ethylene to 1.10 ppm. In sealed BRIC PDFUs, lack of gaseous exchange causes extreme hypoxia ($O_2 < 1.8\%$) and toxic ethylene accumulation (> 3.80 ppm, Fig. 10g).

Table 4 | Ventilation efficiency & biosecurity.

Hardware	Gravity	ε_a	Biosecurity
VEGGIE	1.0g Low	22.5%	Direct cabin exhaust
	0.0g Low	14.2%	Mold risk (52.8% stag.)
	0.0g High	26.2%	Cabin spore dispersion
APH	1.0g Nom	45.8%	Closed loop HEPA
	0.0g Nom	45.0%	Uniform upward sweep
	0.0g High	47.3%	Near-ideal displacement
CHROMEX	0.0g AES	32.3%	Closed (0% export)
	0.0g Sealed	0.0%	Sealed Lexan
CARA Dish	0.0g Draft	18.5%	Contained (Tape)
BRIC-LED	0.0g Sealed	0.0%	Hermetic PDFU

Table 5 | Fan failure resilience & hypoxia.

Hardware	Gravity	t_{Hypoxia}	Resilience
VEGGIE	1.0g	22.0 min	High (chimney updraft)
	0.0g	7.2 min	Critical (stagnation)
APH	1.0g	28.0 min	High (large volume)
	0.0g	8.9 min	Moderate-Low
CHROMEX	0.0g	3.5 min	Extremely Critical
CARA (+L)	0.0g	14.0 min	Moderate (Tape Buffer)
BRIC-LED	0.0g	2.8 min	Critical (Thermal Pocket)

Table 6 | Science sample carrier gas-exchange & microenvironmental metrics.

Carrier	Geometry	Lighting	Gravity	δ_{ext}	r_{tot}	O_2 (%)	C_2H_4	Condensation
CARA	Square (100×100)	1. Light	Earth (1.0g)	4.8	1220	18.4%	0.32 ppm	18.5 h
CARA	Square (100×100)	1. Light	Moon (0.166g)	6.2	1310	16.8%	0.52 ppm	12.0 h
CARA	Square (100×100)	1. Light	0.0g	8.5	1480	14.2%	**0.85 ppm**	**6.5 h**
CARA	Square (100×100)	• Dark	0.0g	9.8	1520	12.5%	**1.10 ppm**	14.0 h
BRIC-LED	Round ($\varnothing 60$)	1. Light	0.0g	20.0	$> 101\text{k}$	**1.8%**	**3.80 ppm**	**1.5 h**
BRIC	Round ($\varnothing 60$)	• Dark	0.0g	25.0	$> 102\text{k}$	**1.2%**	**4.50 ppm**	2.8 h

Methods & Numerical Framework

OpenFOAM Finite-Volume Solver Settings

Simulations were executed within OpenFOAM v2606 using finite-volume discretization of Low-Mach compressible Navier-Stokes equations:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0$$

$$\frac{\partial (\rho \boldsymbol{u})}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \boldsymbol{u}) = -\nabla p_{\text{rgh}} + \boldsymbol{g}\rho + \nabla \cdot \boldsymbol{\tau}_{\text{eff}} + \boldsymbol{S}_m$$

$$\frac{\partial (\rho h)}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} h) = \nabla \cdot (\alpha_{\text{eff}} \nabla h) + S_h$$

Turbulence was modeled using k-ω SST (Menter 1994) with near-wall prism layers ($y^+ \approx 1 - 5$). Vegetative canopies were parameterized as porous media via Darcy-Forchheimer drag sinks:

$$\boldsymbol{S}_m = -\rho \left(\frac{\mu}{K} \boldsymbol{u} + \frac{1}{2} C_d a_v \, |\boldsymbol{u}| \, \boldsymbol{u} \right)$$

where $a_v = \frac{\text{LAI}}{h_c}$ is Leaf Area Density and $C_d = 0.2$. In BRIC-LED, transient 4h ON / 4h OFF diurnal cycles were resolved via conjugate heat transfer coupling LED surface flux ($q'' = 35 \text{ W/m}^2$) and moisture phase change ($\dot{m}_{\text{cond}} h_{fg}$).

Interactive 3D WebGL Dashboard & Multimedia

Interactive WebGL 3D visualizations, animated 4D simulations, and mesh dictionaries are openly accessible:
- **Live Web Portal**: https://dr-richard-barker.github.io/spaceflight-plant-hardware-cfd/
- **Interactive 3D Web Explorer**: `docs/explorer.html`
- **Open-Source Code Repository**: https://github.com/dr-richard-barker/spaceflight-plant-hardware-cfd

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